

Dan Nguyen

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EDUCATION

Gettysburg College

B.S. in Computer Science and Mathematics — GPA: 4.11

- Honors: Phi Beta Kappa, Earl E. Ziegler Junior Mathematics Award, Pi Mu Epsilon

Gettysburg, PA

Jan. 2023 – May 2026

EXPERIENCE

AI Engineer Intern

KPMG

June 2025 – Aug. 2025

Berlin, DE

- Pioneered a project to train a specialized LLM for Global Audit, building an ETL + GraphRAG pipeline with **Python** and **Neo4j** with support of **GitHub Copilot** that improved QA processing efficiency by 30%.
- Engineered prompt-based LLM solutions using **LangChain** and **Jupyter Notebooks** for NLP tasks like markdown extraction, enhancing document parsing reliability by 30%.
- Developed QA dataset generators and an "LLM-as-a-judge" evaluation framework in **Python**, enabling scalable data augmentation and model validation.

Data Scientist Intern

KPMG

May 2024 – Aug. 2024

Berlin, DE

- Leveraged **Python**, **Jupyter Notebooks**, and **MS Excel** to conduct data analysis on public bank datasets, developing a deterministic rule-based baseline model for peer group clustering.
- Engineered a **stochastic unsupervised algorithm** for the **Clara 2.0** AI, a system for **anomaly detection** in accounting data to identify potential failure risks, achieving a **15% improvement** in peer group cohesion.
- Designed and evaluated various peer group quality measures by comparing different methodologies to validate and select the most effective models for the risk assessment tool.

PROJECTS

APPC Archive Assistant | *Django, React, PostgreSQL (pgvector), Docker* | Live Site

Jan. 2026 – Present

- Engineered an end-to-end **Retrieval-Augmented Generation (RAG)** pipeline using **Django** to enable semantic search and intelligent document retrieval across decades of committee archives with **GitHub Copilot**.
- Implemented a **self-querying retriever** using **LlamaIndex** and **Pydantic** to extract structured metadata filters from natural language queries, increasing retrieval precision for specific course codes and date ranges.
- Architected a responsive **React** chat interface featuring **interactive source citations** that link AI-generated insights directly to original PDF documents, ensuring data transparency and model explainability.

Palamedes: Fraternity Management Platform | *Django, PostgreSQL, AWS* | Live Site

Dec 2025 – Present

- Engineered a full-stack, multi-tenant web application using **Django**, **PostgreSQL**, and **AWS S3** to streamline chapter operations, including automated member onboarding and activity tracking.
- Implemented a robust **Role-Based Access Control (RBAC)** system to dynamically assign granular user permissions and secure hierarchical approval workflows for administrative tasks.
- Architected a comprehensive financial dashboard integrated with the **Stripe API**, enabling executive officers to query directories, issue bulk charges, and automatically resolve member balances via asynchronous webhooks.

Mancala AI Player | *Java, Python* | View on GitHub

Jan 2025 – Feb 2025

- Implemented a competitive Mancala AI using minimax search with alpha-beta pruning, transposition tables, and heuristic evaluation for efficient state-space exploration.
- Designed a dynamic time management module to adapt search depth and node expansion based on real-time computational budget and board complexity, achieving top-tier class performance.

TECHNICAL SKILLS

Languages & Frameworks: Java, Python, JavaScript, HTML/CSS, PHP; Django, Bootstrap, LangChain, LangGraph, scikit-learn

Data Stack: PostgreSQL, MySQL, SQL, Neo4j; pandas, NumPy, Matplotlib, Microsoft Excel

Developer Tools: Git, Docker, Azure DevOps, AWS (S3), Render